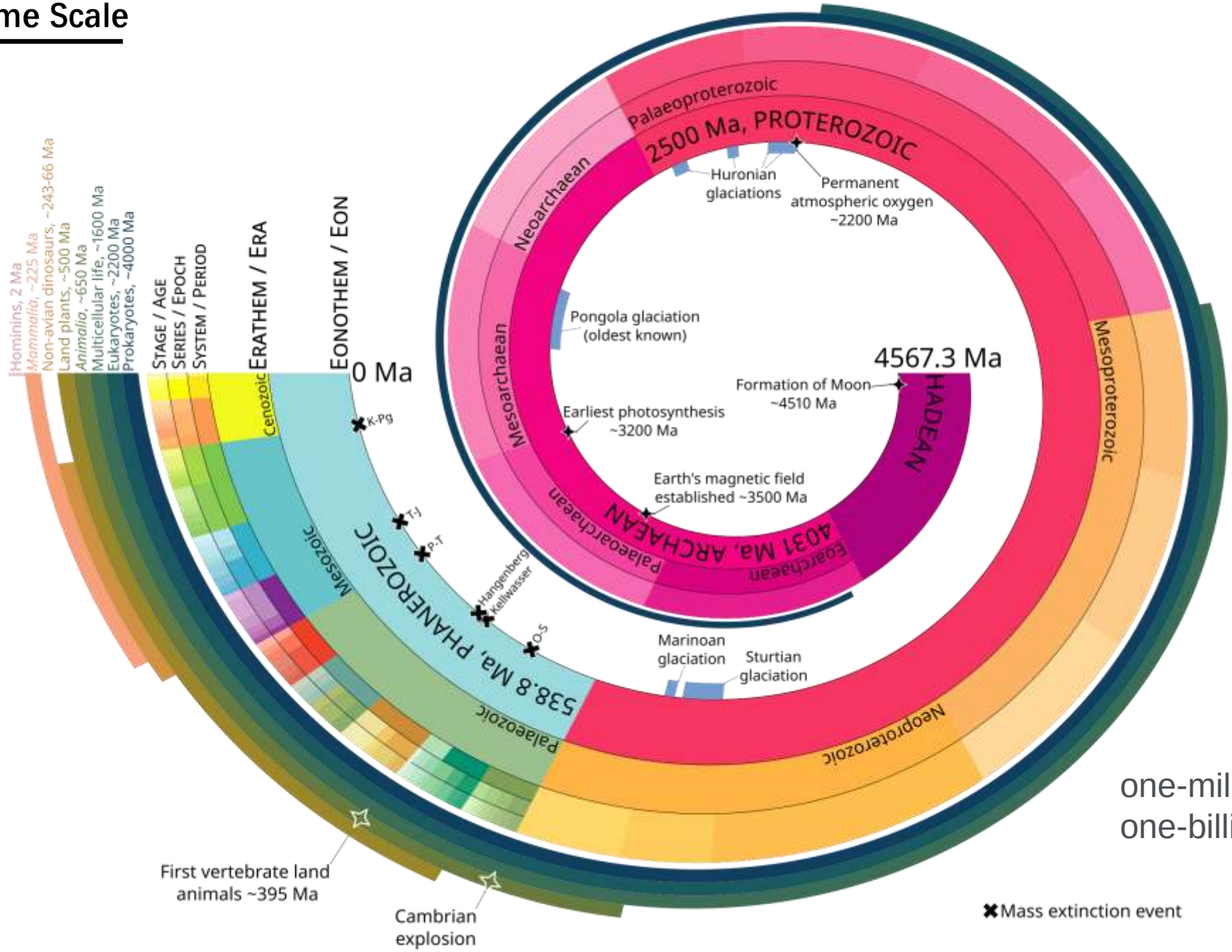
GEOLOGICAL TIME SCALE2

Sophia

2024/06/18

Geological Time Scale

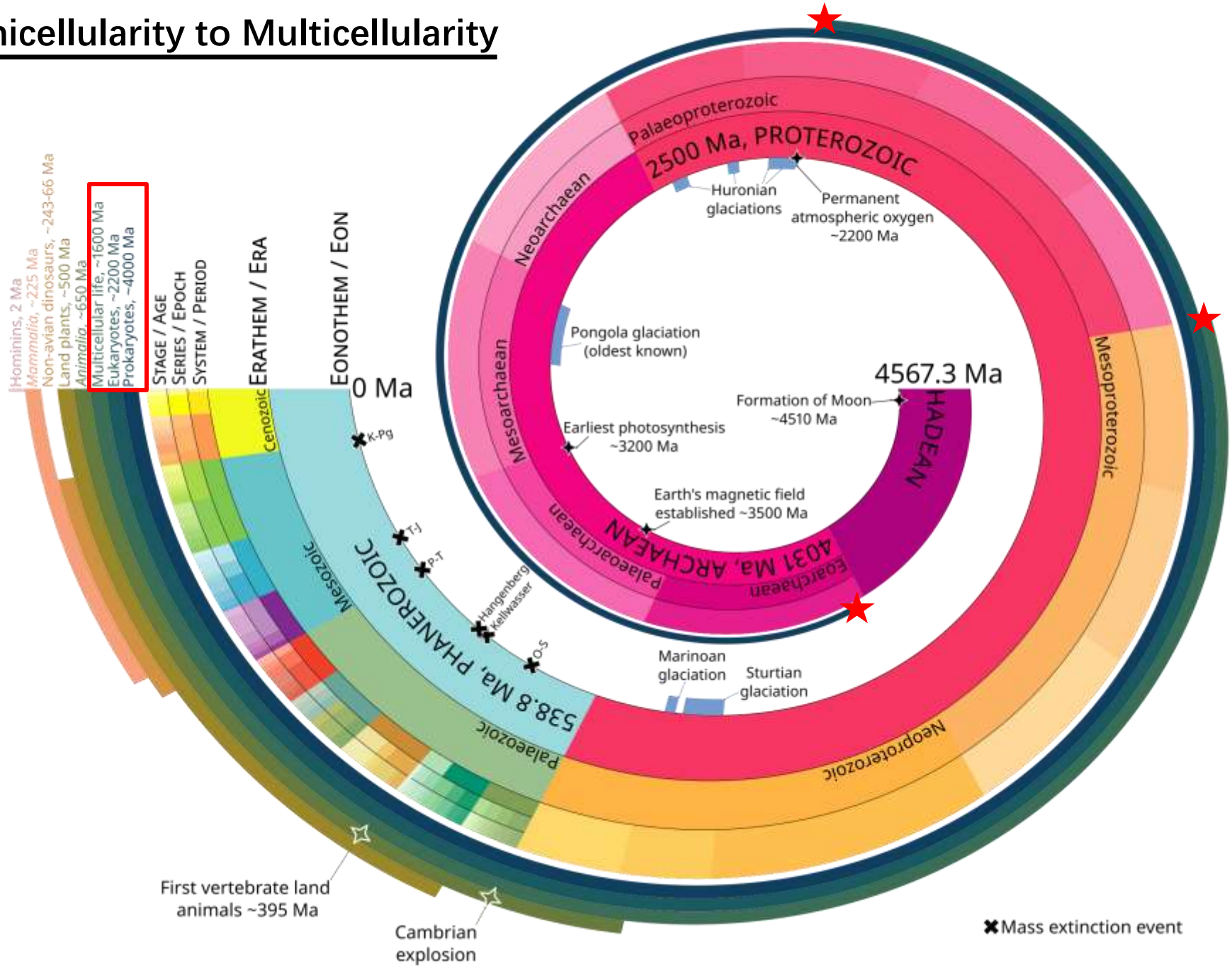


one-million years = 1Ma
one-billion years = 1Ga

QUESTIONS TO BE ANSWERED

1. When did the Multicellularity Evolved From Unicellularity?
2. When did phytoplankton first evolve?
3. What shaped phytoplankton's speciation?

Q1: From Unicellularity to Multicellularity



PHYTOPLANKTON

- What is phytoplankton?
- When did phytoplankton first evolve?

REVIEW

The Evolution of Modern Eukaryotic Phytoplankton

Paul G. Falkowski,[1,2*] Miriam E. Katz,[2] Andrew H. Knoll,[3] Antonietta Quigg,[1†] John A. Raven,[4] Oscar Schofield,[1] F. J. R. Taylor[5]

Falkowski et al., 2004

What are phytoplankton?

phytoplankton = *phyto* (plant) + *plankton* (made to wander or drift)

They are **microscopic organisms that live in watery environments**, both marine and fresh waters.

Phytoplankton are extremely diverse, varying from photosynthesizing bacteria (cyanobacteria), to plant-like diatoms, to armor-plated coccolithophores (drawings not to scale). (Collage adapted from drawings and micrographs by Sally Bensusen, NASA EOS Project Science Office.)

Classification and phylogenetic distribution

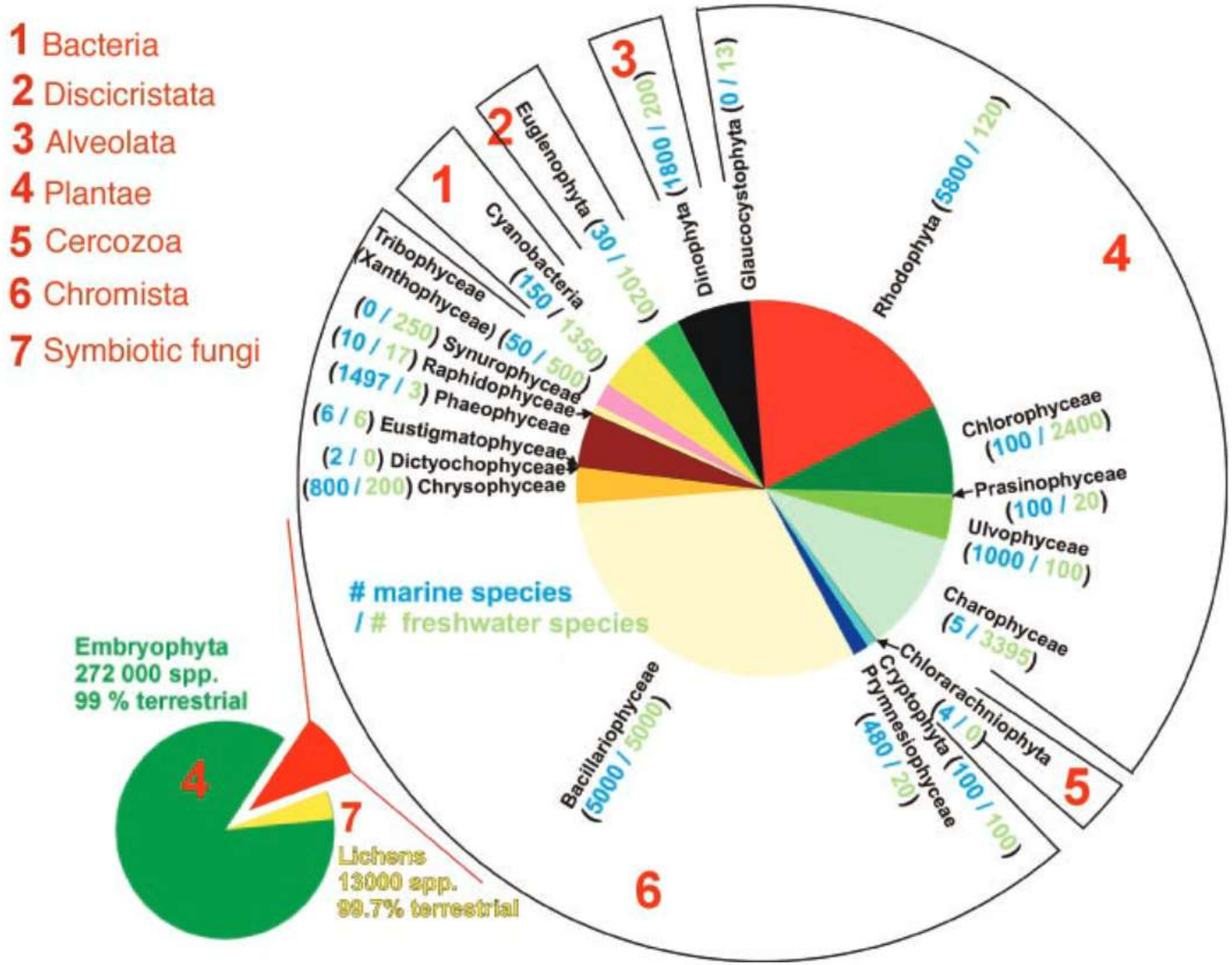


Fig. A phylogenetic distribution of terrestrial and aquatic photoautotrophs based on morphological characteristics. (Falkowski et al., 2004)

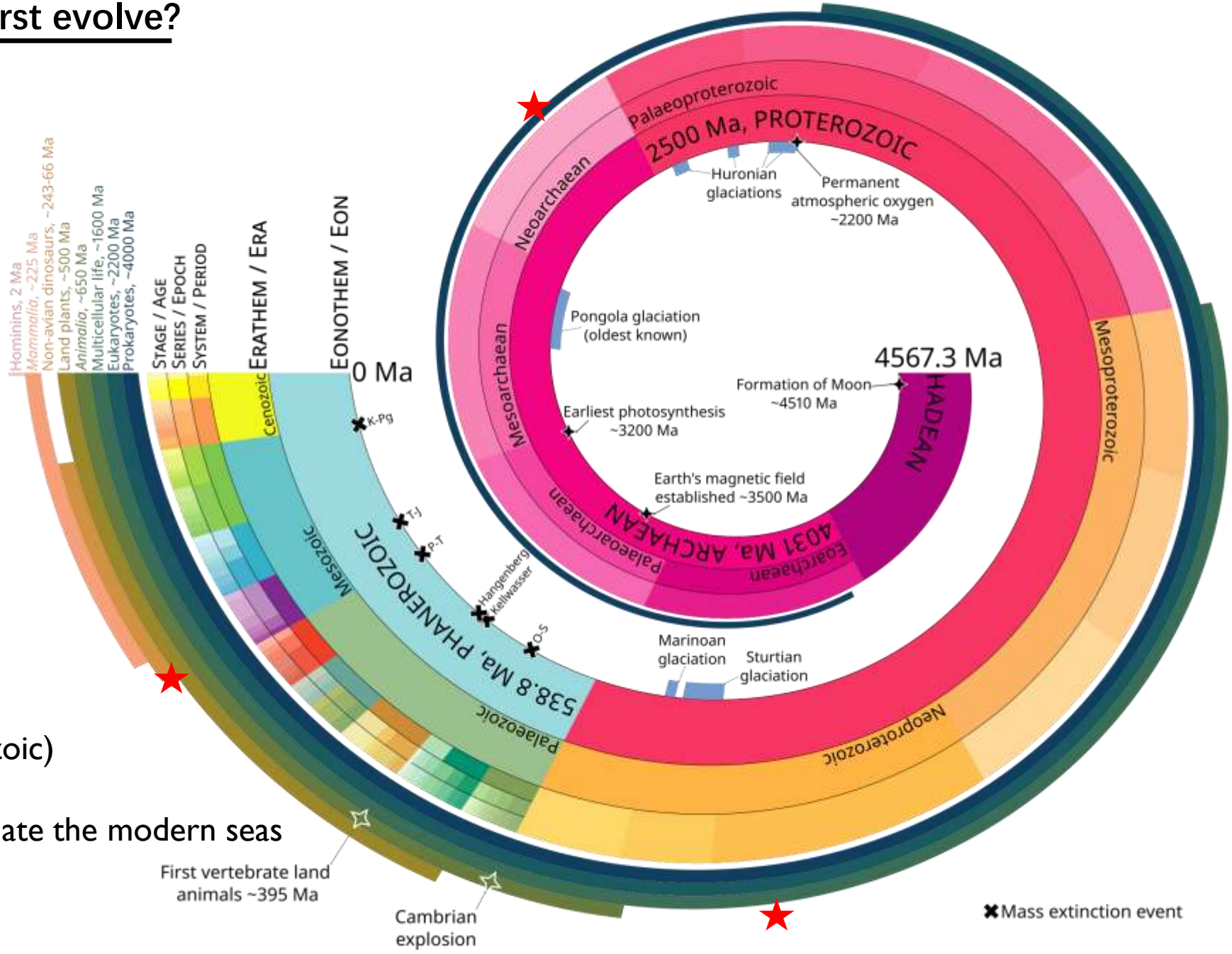
When did phytoplankton first evolve?

From fossil evidence

- Prokaryotes:
  ★2.7 Ga (Archaean)

- Eukaryotes
  ★1.6 ~ 1.8 Ga ago (Proterozoic)

Principal phytoplankton dominate the modern seas
★251 ~ 65 Ma (Mesozoic)



PHYTOPLANKTON

- What shaped their evolution and speciation?